

Lending Under Imperfect Information

Discrimination, Credit Allocation, and Machine Learning

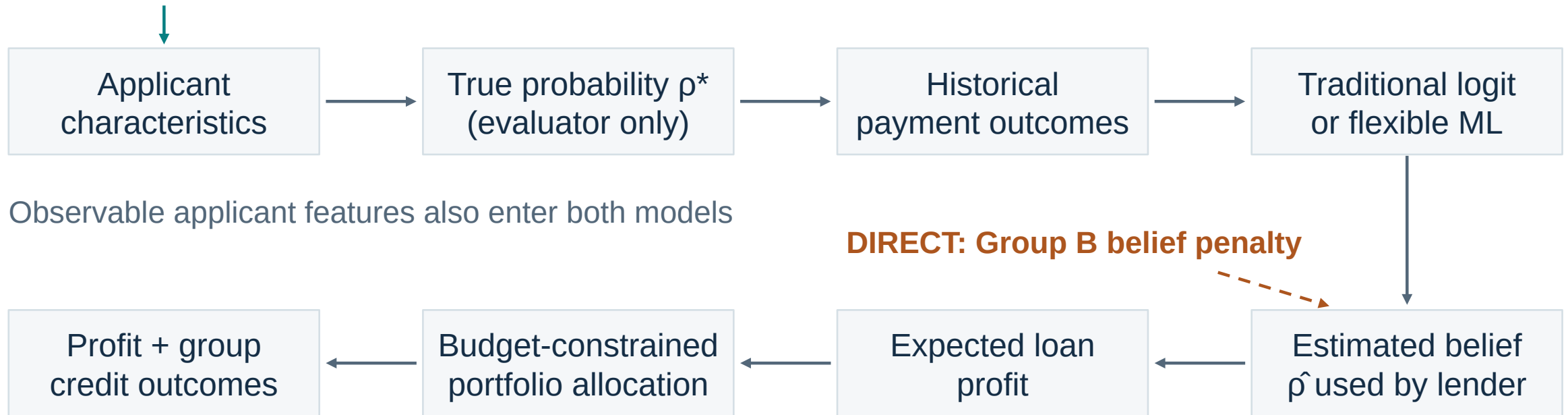
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How do imperfect estimates of repayment risk and different discrimination mechanisms affect lending decisions across groups, and what changes when the bank uses machine learning?

Prediction → Profit → Credit Allocation

Overall Research Model

SYSTEMIC: earlier opportunity access



$\rho = P(\text{next payment} \mid \text{all prior payments made})$

Expected profit = $\sum_{t=1}^T (\text{payment}_t \times \hat{\rho}^t) - \text{principal} - \text{cost}$

From Approval Analysis to Economic Lending

VERSION 1

Approval disparity benchmark

VERSION 2

Risk → profit → constrained allocation

Population + accounting

Synthetic applicants
Repayment and default paths
Survival-weighted cash flows

Risk estimation

Traditional logistic lender
Flexible ML lender
Hidden-truth evaluation

Allocation + mechanisms

Binary budget optimization
Direct belief distortion
Upstream opportunity process

10,000 applicants 6k / 2k / 2k cohorts 120-period stylized contract

284 tests passed (documented) • Repeated-seed analysis ongoing

Source: project status and Version 2 methodology; latest documented test result, not a new test run.

Prediction Accuracy and Economic Value

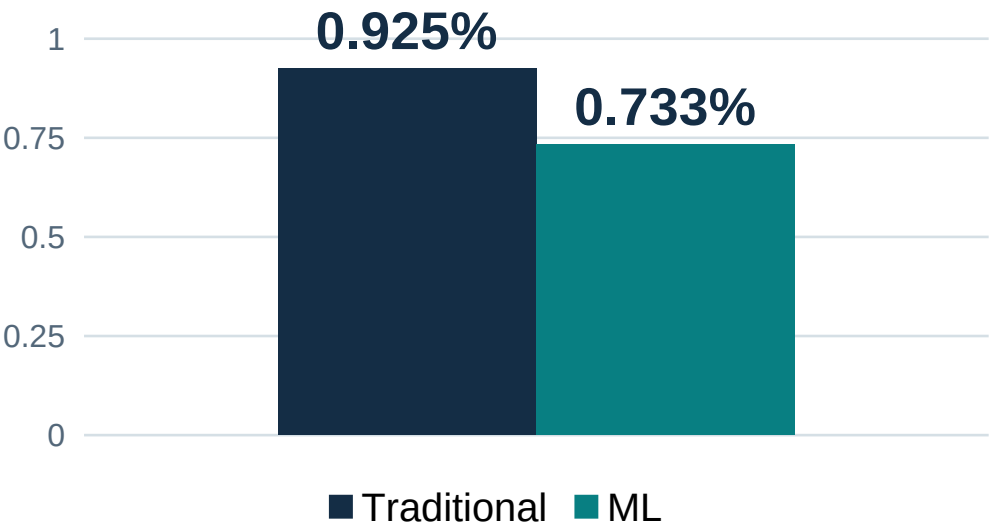
Probability and request-profit errors

	Traditional	ML (HistGB)
Additive risk MAE	0.000364	0.001341
Nonlinear risk MAE	0.013471	0.018465
Nonlinear profit MAE	\$13,297	\$12,026

Risk MAE is in probability units. Lower is better.

Nonlinear world, 20% budget

Regret as % of oracle expected profit



Traditional had lower regret in the other five primary world/budget comparisons.

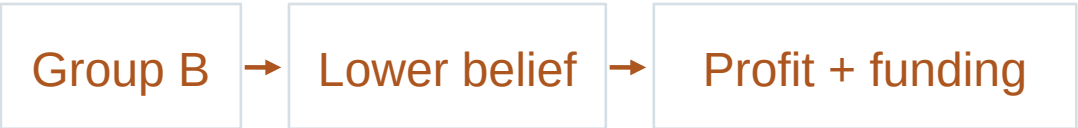
Economic value depends on functional form and capital scarcity.

Source: held-out Version 2 synthetic benchmarks; one development population, not a general model ranking.

Direct and Systemic Mechanisms

DIRECT BELIEF DISTORTION

$\delta = 0.20$; true borrower risk unchanged



Additive world, traditional lender, 40% budget

+3.96 pp

Group A funding

−3.85 pp

Group B funding

B−A funding-gap change: −7.80 pp

Separate oracle-reference cost: \$211K

SYSTEMIC OPPORTUNITY

$s = 0.40$; final lender excludes group



70.22% → 61.83%

Group B opportunity access

417 applicants lose access (8.40% of B)

Among switchers: income −\$8.3K
assets −\$9.7K; employment −1.48 years

Current Status and Next Steps

COMPLETED

Economic simulation and risk models
Budget-constrained portfolios
Direct and systemic experiments

CORRECTED MINI-BATCH

5 replications / 360 portfolios

Validated; the 50-run study is unfinished.
19 legacy successes excluded as provisional.

284 passed / 32 existing warnings

NEXT

- 01 Authorize 45 remaining corrected runs
- 02 Combine direct + systemic mechanisms
- 03 Complete robustness analysis
- 04 Update dashboard; write final paper

Separate estimation error from direct and upstream discrimination.